**HABIB UNIVERSITY**

**DHANANI SCHOOL OF SCIENCE AND ENGINEERING**

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| EE251/CE252 | Spring'26 | HW 1<br>SOLUTION | Marks 50<br>Weight 2.5% | Due: As announced |

**Important Notes**

Preceding the solutions, you must include the following information:

- Your name
- Student ID
- Course ID
- Homework number
- Date and time submitted

Solution should be elaborative, sketch neat circuits wherever required, show clearly all steps of calculation & mention reference (for example formula, law, rule) to strengthen your understanding to solve the electric circuit problems.

*Also, please comply with Academic Integrity Policies of Habib University related to plagiarism while submitting this assignment for grading.*

**Late Submission Policy**

As announced in course syllabus.

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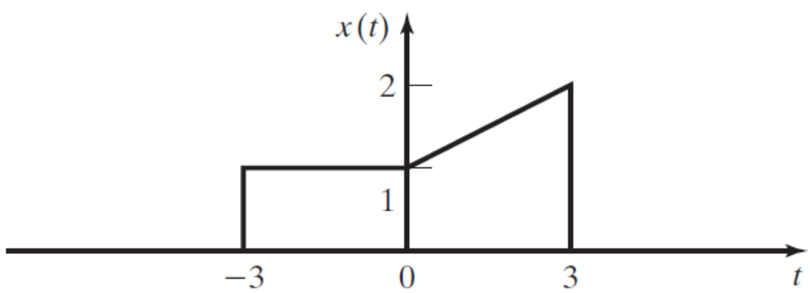
| **CLO-1** | Apply knowledge of signals and systems to categorize and perform basic operations on signals and systems. | PLO-1 (Engineering Knowledge) | Cognitive Level-3 (Applying) |

## Problem 1

Consider the signal shown below




Plot the following

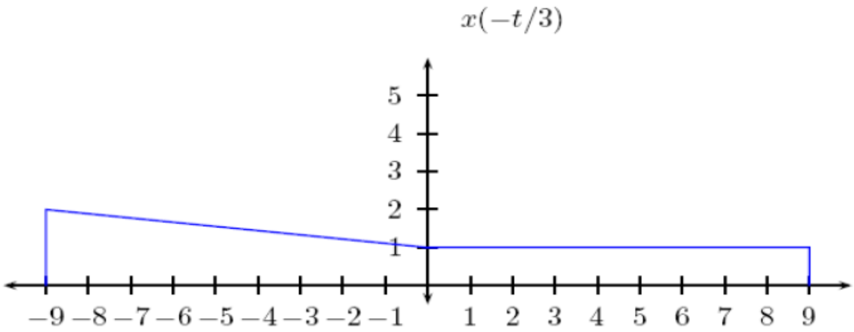
**(i)** $x(-t/3)$ **(ii)** $x(-t)$

**(iii)** $x(3 + t)$ **(iv)** $x(2 - t)$

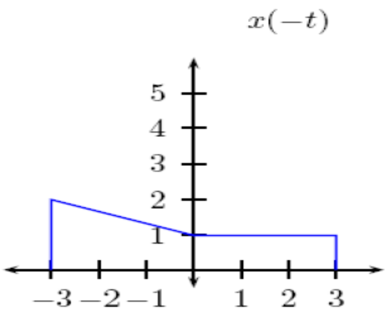
Solution:

(a)

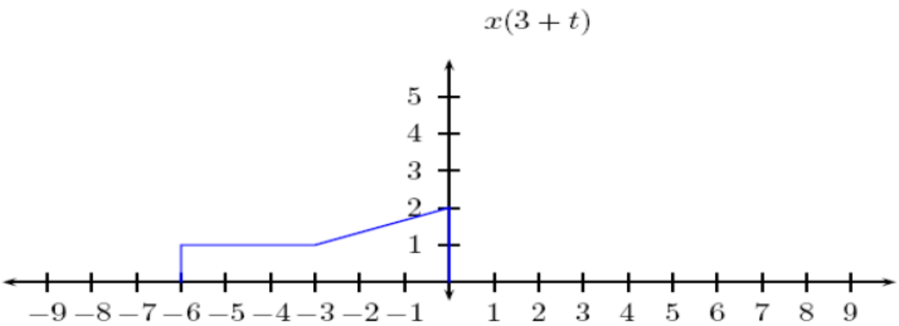
(i)



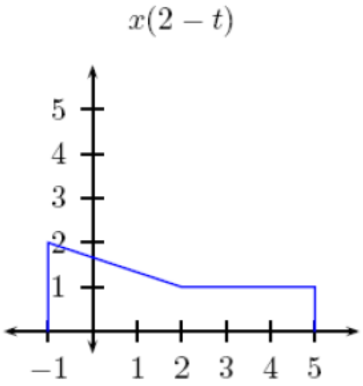

(ii)




(iii)



(iv)




## Problem 2

For each of the signals given, determine mathematically whether the signal is even, odd, or neither. Sketch the waveforms to verify your results.

**(a)** $x(t) = -4t$

**(b)** $x(t) = e^{-|t|}$

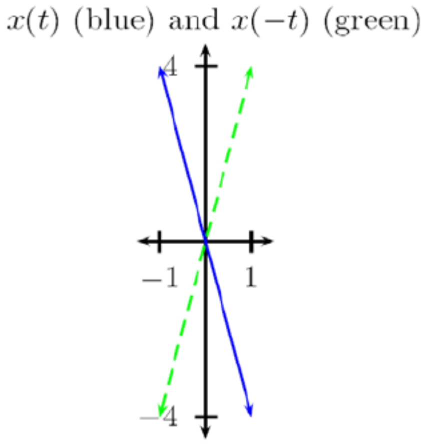
**(c)** $x(t) = 5\cos 3t$

**(d)** $x(t) = \sin(3t - \frac{\pi}{2})$

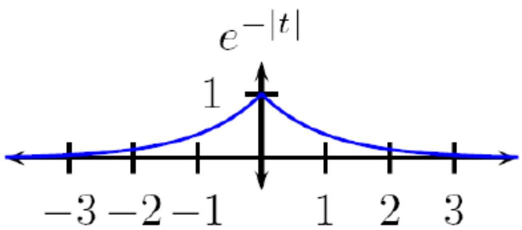
**(e)** $x(t) = u(t)$

Solution:

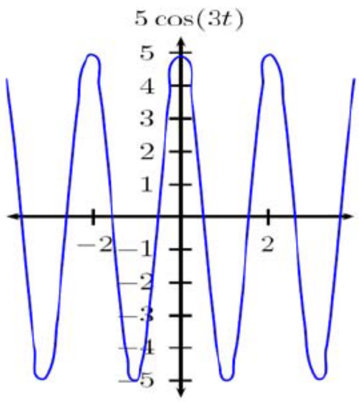
a) $-4t = -(-4(-t))$ so it is odd.



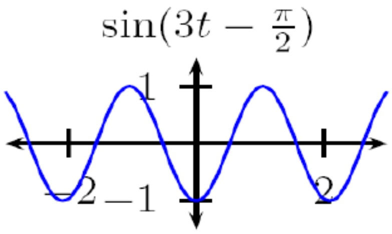

b) $e^{-|t|} = e^{-|-t|}$ so it is even ($|t| = |-t|$).



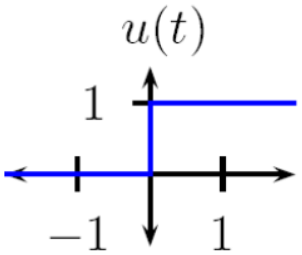

c) Since $\cos(t)$ is even, $5\cos(3t)$ is also even.



d) $\sin(3t - \frac{\pi}{2}) = -\cos(3t)$ which is even:




e) $u(t)$ is neither even nor odd; for example $u(3) = 1$ but $u(-3) = 0 \neq -u(3), \; \neq u(3)$.




## Problem 3

Give proofs of the following statements:

**(a)** The sum of two even functions is even.

**(b)** The sum of two odd functions is odd.

**(c)** The sum of an even function and an odd function is neither even nor odd.

Solution:

(a) Let $z(t)$ be the sum of two even functions $x_1(t)$ and $x_2(t)$. To show that $z(t)$ is even, we need to show that $z(t) = z(-t)$ for all $t$. This is easy to show, since $z(t) = x_1(t) + x_2(t)$ and $z(-t) = x_1(-t) + x_2(-t)$ (since to get $z(-t)$ we just plug in $-t$ everywhere for $t$, which amounts to just plugging in $-t$ in $x_1(t)$ and $x_2(t)$). Now since $x_1(t)$ and $x_2(t)$ are even, by definition $x_1(t) = x_1(-t)$ and $x_2(t) = x_2(-t)$ so $x_1(t) + x_2(t) = x_1(-t) + x_2(-t)$ so $z(t) = z(-t)$.

(b) Let $x_1(t)$ and $x_2(t)$ be two odd functions. Then $x_1(-t) + x_2(-t) = -x_1(t) + (-x_2(t)) = -(x_1(t) + x_2(t))$ which shows that $x_1(t) + x_2(t)$ is odd.

(c) Let $z(t) = x_1(t) + x_2(t)$ as in part a, where now $x_1(-t) = x_1(t)$ and $x_2(-t) = -x_2(t)$. We need to show that $z(t) \neq z(-t)$, $z(t) \neq -z(-t)$. Consider that $z(-t) = x_1(-t) + x_2(-t) = x_1(t) - x_2(t)$. In order to have $z(t)$ be even, we would therefore need to have $x_1(t) + x_2(t) = x_1(t) - x_2(t)$ for all $t$, which is equivalent to having $x_2(t) = -x_2(t)$ for all $t$, which is not possible for nonzero $x_2(t)$. Similarly, in order to have $z(t)$ be odd, we would need to have $z(t) = -z(t) \implies x_1(t) + x_2(t) = x_2(t) - x_1(t)$, which is not possible for nonzero $x_1(t)$. So the sum of an even and odd function must be neither even nor odd.

## Problem 4

Prove mathematically that the signals given below are periodic. Also find each signal's fundamental period.

**(a)** $x(t) = 7\sin 3t$

**(b)** $x(t) = \sin(8t + 30^\circ)$

**(c)** $x(t) = e^{j2t}$

**(d)** $x(t) = \cos t + \sin 2t$

**(e)** $x(t) = e^{j(5t+\pi)}$

**(f)** $x(t) = e^{-j10t} + e^{j15t}$

Solution:

(a) $\sin(t) = \sin(t + n2\pi)$ for any integer $n$, so $7\sin(3t) = 7\sin(3t + n2\pi) = 7\sin\left(3(t + n\frac{2\pi}{3})\right)$; therefore $x(t)$ is periodic with fundamental period $T_0 = \frac{2\pi}{3}$ and fundamental frequency $\omega_0 = \frac{2\pi}{T_0} = 3$.

(b) $\sin(8(t + \frac{2\pi}{8}) + 30) = \sin(8t + 2\pi + 30) = \sin(8t + 30)$.
$\omega_0 = 8$ and $T_0 = \frac{2\pi}{8} = \frac{\pi}{4}$.

(c) $e^{jt} = \cos(t) + j\sin(t)$ is periodic with fundamental period $2\pi$, so $e^{j2t}$ is periodic with fundamental period $\frac{2\pi}{2} = \pi$, and fundamental frequency $\omega_0 = 2$.

(d) $\cos(t) = \cos(t+n2\pi)$ for any integer $n$, and $\sin(2t) = \sin(2(t+m\pi))$ for any integer $m$, so $\cos(t)+\sin(2t)$ will be periodic with period $T_0$ if $\cos(t) + \sin(2t) = \cos(t + T_0) + \sin(2(t + T_0))$. This will hold as long as $T_0 = n2\pi$ and $T_0 = m\pi$ for some integers $n$ and $m$, and the fundamental period is the smallest value for which this holds, which is $T_0 = 2\pi$, with fundamental frequency $\omega_0 = 1$.

(e) $e^{j(5t+\pi)} = e^{j\pi}e^{j5t}$. So the phase shift of $\pi$ just means a complex constant (constant with respect to time) out front and does not effect periodicity of the signal $e^{j5t}$, which has fundamental period $T_0 = \frac{2\pi}{5}$ and $\omega_0 = 5$.

(f) $e^{-j10t}$ and $e^{j15t}$ are both periodic with periods $\frac{\pi}{5}, \frac{2\pi}{15}$ and their sum is periodic with period $T_0 = LCM(\frac{\pi}{5}, \frac{2\pi}{15}) = \frac{2\pi}{5}$ and $\omega_0 = 5$:
$e^{-j10(t+\frac{2\pi}{5})} + e^{j15(t+\frac{2\pi}{5})} = e^{-j10t}e^{-j4\pi} + e^{j15t}e^{j6\pi}$ and since $e^{-j4\pi} = 1$ and $e^{j6\pi} = 1$ this $= e^{-j10t} + e^{j15t}$.

## Problem 5

Evaluate the following integrals

**(i)** $\displaystyle\int_{-\infty}^{\infty} \cos(2t)\delta(t)dt$

**(ii)** $\displaystyle\int_{-\infty}^{\infty} \sin(2t)\delta(t - {}^{\pi}/_{4})dt$

**(iii)** $\displaystyle\int_{-\infty}^{\infty} \cos[2(t - {}^{\pi}/_{4})]\delta(t - {}^{\pi}/_{4})dt$

**(iv)** $\displaystyle\int_{-\infty}^{\infty} \sin[(t - 1)]\delta(t - 2)dt$

**(v)** $\displaystyle\int_{-\infty}^{\infty} \sin[(t - 1)]\delta(2t - 4)dt$

Solution:

Recall the rules about integrating delta functions: $\delta(t)$ is nonzero only at $t = 0$, so $x(t)\delta(t) = x(0)\delta(t)$, and $\int_{-\infty}^{\infty}\delta(t)dt = 1$, so $\int_{-\infty}^{\infty}x(t)\delta(t)dt = \int_{-\infty}^{\infty}x(0)\delta(t)dt = x(0)\int_{-\infty}^{\infty}\delta(t)dt = x(0)$. We can time-shift the delta function: $\delta(t-t_0)$ is nonzero only at $t = t_0$, so $x(t)\delta(t-t_0) = x(t_0)\delta(t-t_0)$ and $\int_{-\infty}^{\infty}x(t)\delta(t-t_0)dt = x(t_0)$.

i) $\int_{-\infty}^{\infty}\cos(2t)\delta(t)dt = \cos(2\cdot 0)\int_{-\infty}^{\infty}\delta(t)dt = 1$.

ii) $\delta(t-\frac{\pi}{4})$ is a time-shifted version of $\delta(t)$, and is nonzero only at $t = \frac{\pi}{4}$. So:

$$\begin{aligned}\int_{-\infty}^{\infty}\sin(2t)\delta(t-\frac{\pi}{4})dt = &\quad \int_{-\infty}^{\infty}\sin(2\cdot\tfrac{\pi}{4})\delta(t-\tfrac{\pi}{4})dt\\ = &\quad \sin(\tfrac{\pi}{2})\int_{-\infty}^{\infty}\delta(t-\tfrac{\pi}{4})dt = \sin(\tfrac{\pi}{2}) = 1\end{aligned}$$

iii) $\cos(2(t-\frac{\pi}{4}))\delta(t-\frac{\pi}{4}) = \cos\left(2(\frac{\pi}{4}-\frac{\pi}{4})\right)\delta(t-\frac{\pi}{4}) = 1\cdot\delta(t-\frac{\pi}{4})$, so the integral of this is 1.

iv) $\delta(t-2)$ is nonzero only at $t = 2$. Therefore $\int_{-\infty}^{\infty}\sin((t-1))\,\delta(t-2)dt = \sin(2-1) = \sin(1) = 0.8414....$

v) $\delta(2t-4)$ is nonzero at $2t-4 = 0 \implies t = 2$. So:

$$\int_{-\infty}^{\infty}\sin(t-1)\,\delta(2t-4)dt = \sin(2-1)\int_{-\infty}^{\infty}\delta(2t-4)dt$$

To figure out the integral, we can change variables—let $u = 2t$, so $dt = \frac{du}{2}$ and the $-\infty, \infty$ limits stay the same. This gives: $\int_{-\infty}^{\infty}\delta(2t-4)dt = \int_{-\infty}^{\infty}\delta(u-4)\frac{du}{2} = \frac{1}{2}$, so we get:

$$\int_{-\infty}^{\infty}\sin(t-1)\,\delta(2t-4)dt = 0.5\sin(1) = 0.4207...$$